

Robin CR

Chennai, Tamil Nadu

☎ +91 8848667963 | ✉ robinchovallurraju@gmail.com | [in linkedin.com/in/robin-cr/](https://www.linkedin.com/in/robin-cr/)
🐙 github.com/robin005cr

Personal Profile

4+ years experienced Embedded Software Developer with proven success in building algorithms. Highly adept at C++ application development, firmware development, ADAS, Design Patterns and multi-threading. Passionate engineer with the ability to apply techniques and algorithm development to solve real world problem.

Work Experience

Automotive Robotics India (ARi)

Chennai, Tamil Nadu

Senior Software Engineer

January 2025 – Present

Client: CATERPILLAR (Onsite)

LIDAR Perception for Surface Hauling truck

- Engineered Moving Obstacle Detection solutions leveraging lidar sensor.
- Built and optimized Range image-to-point cloud converters, enabling support for multiple lidar types.
- Created and maintained efficient parsers for ROS node communication.
- Integrated open-source Point Cloud libraries into C++ projects to accelerate development.
- Worked in Path planning, Sensor fusion and SLAM algorithms.
- Drove KPI improvements by investigating and resolving root causes of KPI mismatches.
- Developed robust backend systems for a playback tool.

Bosch Global Software Technologies

Coimbatore, Tamil Nadu

Embedded Software Engineer

August 2022 – December 2024

Interior Monitoring System (IMS) Project – Camera Perception

- Developed algorithms for driver drowsiness module in C++ following design patterns and clean code.
- Able to handle real world bugs which require proper strategies, Problem Solving and analytical skills.
- Developed internal tools for simulating driver behaviors for debugging the issues without hardware.
- A proper research and contributor for technical problems raised by client.
- Modules implementations such as interfaces, precondition handlers, error handlers, schedulers etc.
- Successfully addressed and resolved Level 8 and 9 QAC++ warnings in the codebase.

Instrument Cluster Project

- Worked in all phases of Agile Scrum methodology (Requirement - Design - Coding - Testing).
- Issue reproduction based on scenarios (Bugs and Change Requests).
- Functionality testing based on all dependent modules.
- Writing unit tests and performing testing to ensure software functionality.
- Implemented an algorithm that will modify the fixed parameters in RTE layer into a configurable in application layer.
- Base level knowledge on Diagnostics, Bootloader and RTE layer is gained.

Gadgeon Systems Pvt Ltd

Kochi, Kerala

Embedded Software Developer

July 2021 - August 2022

- Developed codecs for different sensors using C programming language in a Linux OS environment. (Gateway Programming)
- Developed custom protocols to communicate with different sensors in embedded C.
- Platform dependent and independent code development on different ARM based microcontrollers.
- Interfaced accelerometer, gyroscope, and GPS sensors (Bare metal programming).

Projects

- **Calendar and Clock** : A project which addresses all the problems related to calendar and clock.
- **CPU-GPU Selector** : Implemented a system that decides the model training to be done on CPU, GPU, or GPU on cloud.

Skills

- **Microcontrollers**: ARM Cortex M3 LPC1768, ESP8266
- **Protocols**: UART, I2C, SPI, CAN
- **Design**: Enterprise Architecture, PlantUML
- **Debugger**: Valgrind, GDB, Lauterbach Trace32
- **Automotive testing tools**: CANoe
- **Project Manager**: Gerrit, JIRA
- **Unit Testing**: Gtest, BOOST
- **Version control system**: Git, RTC, BitBucket
- **Build**: Cmake, Sconsript, makefile
- **OS**: Ubuntu
- **AI tools**: Github Copilot, Perplexity, Google AI studio

Competencies

- **Data Structures**
Array, Linked list, STL library.
- **Embedded C programming**
State machine, Bit wise programming, handshaking mechanisms.
- **C++ Application development**
Move Semantics, Smart Pointers, Templates, Functional Programming, Atomics.
- **AI Integration**
Synthetic dataset generation, Local LLM integration

Certifications

Embedded System Professional

Translot, Kochi

- Solid foundation on basics of embedded C programming.
- Completed the projects based on EEPROM and display systems.
- Working knowledge on resource utilization, shared memory, and schedulers.
- Proper understanding of Linux Fundamentals (System calls, Booting, file system) is gained.

Professional C++ and Clean code

Bosch, Coimbatore

- Leveraged the OOPs and SOLID principles with examples.
- Discussed common pitfalls in C++ for embedded software development.
- Covered Design Patterns explaining their roles in code reusability and scalability.
- Clean coding guidelines and commons practices were addressed.

Education

Rajagiri School of Engineering and Technology

Bachelor of Technology in ECE

Completed with a GPA of 8.2

Kochi, Kerala

Aug 2017 - Jul 2021